Ambient-Noise Interferometry Across the Yakima Fold and Thrust Belt: Surface-Wave Modes and Shallow Velocity Heterogeneity

Aaron J. Girard[1], Daniel R. H. O'Connell[2], and Jeffrey Shragge[1]

[1]Colorado School of Mines, [2]Retired, Denver, CO

agirard@mines.edu, geomgcn@gmail.com, jshragge@mines.edu

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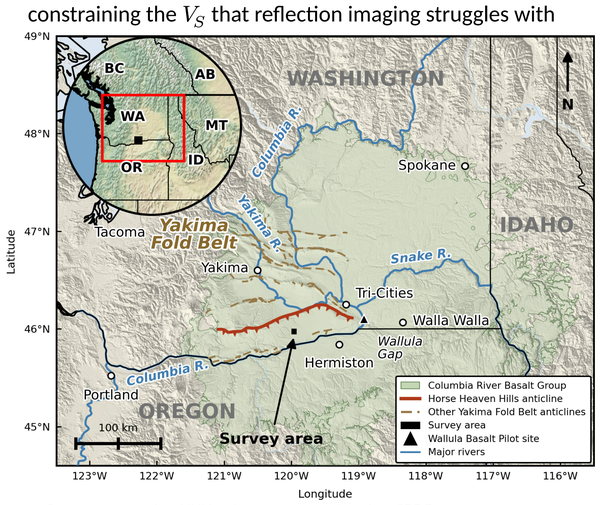
MOTIVATION

- Horse Heaven Hills (HHH): anticlinal ridge of the Yakima Fold and Thrust Belt
- Layered Columbia River Basalt Group (CRBG): dense flow interiors, weathered low-velocity flow tops
- Slow–fast layering favors higher surface-wave modes
- **Project TrapRock**: Carbon Containment Lab and Washington Geological Survey assess the CRBG for carbon storage
- **Low frequencies**: layered basalt scatters high-frequency energy, while long wavelengths reach 1 km depth, constraining the V_S that reflection imaging struggles with

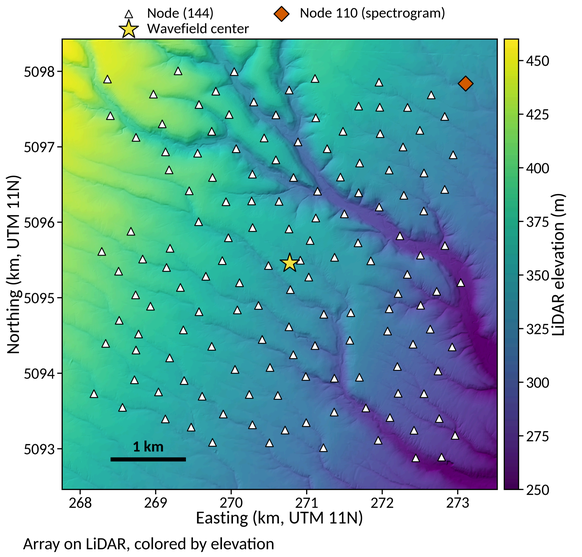


Survey area on the HHH anticline (red), within the CRBG (green)

CONTINUOUS NODAL ARRAY

- **144** three-component (3-C) 2 Hz nodes
- **Seven weeks** continuous, Aug to Oct 2025
- **5.5 km** aperture; **170 m** relief, high NW to low SE
- Passive component of the TrapRock nodal dataset (Girard et al., submitted)

GTI NuSeis D3 node prior to burial



Array on LiDAR, colored by elevation

REFERENCES AND ACKNOWLEDGMENTS

Davis, D., Shragge, J., de Ridder, S., Girard, A. J., & Pandey, A. (2026). Interferometric frequency-domain gradiometry. *Geophysical Journal International*, 245(3), ggag146.

Girard, A. J., & Shragge, J. (2020). Automated processing strategies for ambient seismic data. *Geophysical Prospecting*, 68(1), 293–312.

Girard, A. J., et al. (2024). Observations from the seafloor: Ultra-low-frequency ambient ocean-bottom nodal seismology at the Amendment field. *Geophysical Journal International*, 239(1), 17–36.

Luu, K. (2021). disba: Numba-accelerated computation of surface wave dispersion. doi:10.5281/zenodo.3987395.

Girard, A. J., et al. (submitted). An active- and passive-source nodal seismic dataset for CO_2 sequestration site characterization in the Columbia River Basalts of south-central Washington, U.S.A.

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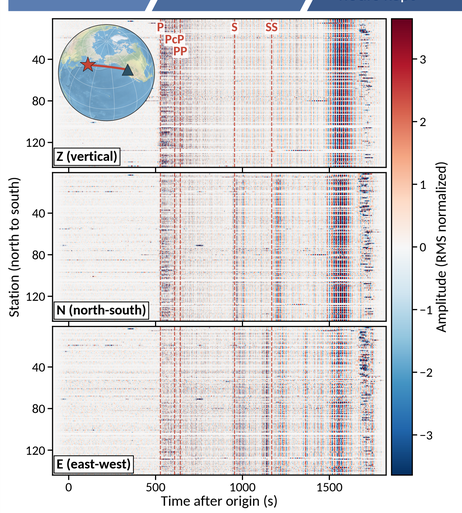
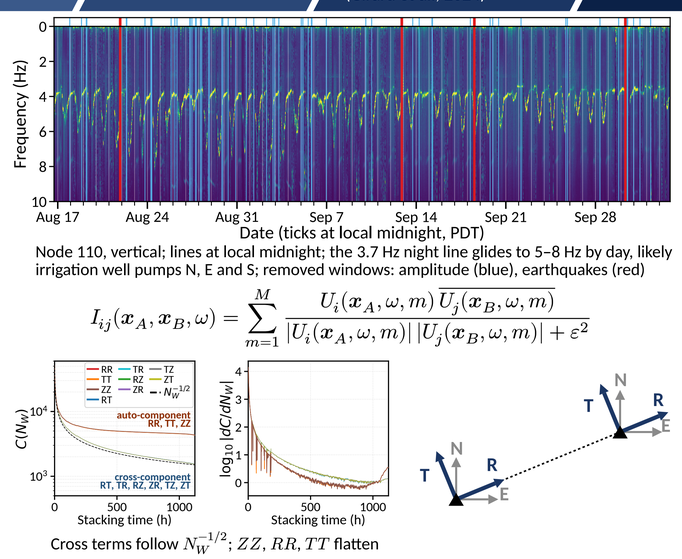
PROCESSING

Decimate anti-alias to 25 Hz → **Detrend** multi-pass spline → **Window selection** amplitude and earthquakes: **1125 of 1200** hours kept → **Temporal balance** Girard & Shragge (2020) → **GPU cross-coherence** with node locations, rotates N–E to R–T in one pass (Girard et al., 2024) → **9-C VSGs**

virtual shot gathers, 144 × 144 nodes



M_w 7.8 Kamchatka aftershock, 18 Sep 2025: coherent on all 144 stations and removed before stacking



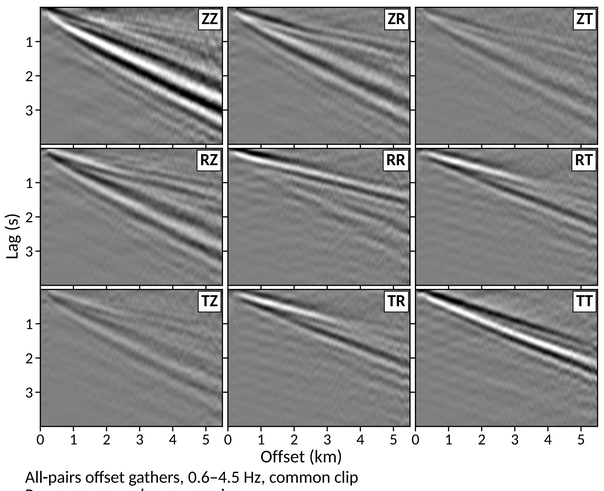
Node 110, vertical; lines at local midnight; the 3.7 Hz night line glides to 5–8 Hz by day, likely irrigation well pumps N, E and S; removed windows: amplitude (blue), earthquakes (red)

$$I_{ij}(\boldsymbol{x}_A, \boldsymbol{x}_B, \omega) = \sum_{m=1}^{M} \frac{U_i(\boldsymbol{x}_A, \omega, m)\, \overline{U_j(\boldsymbol{x}_B, \omega, m)}}{|U_i(\boldsymbol{x}_A, \omega, m)|\, |U_j(\boldsymbol{x}_B, \omega, m)| + \varepsilon^2}$$

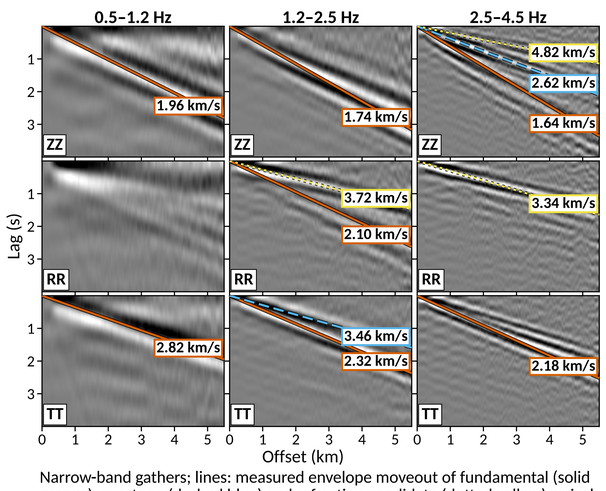
Cross terms follow $N_W^{-1/2}$; ZZ, RR, TT flatten

NINE-COMPONENT VSGs

- Surface-wave-dominated field
- ZZ, RR (Rayleigh) and TT (Love): dispersive moveout
- ZR, RZ: energetic, reciprocity-linked
- ZT, TZ, RT, TR: weak but nonzero, indicating heterogeneity



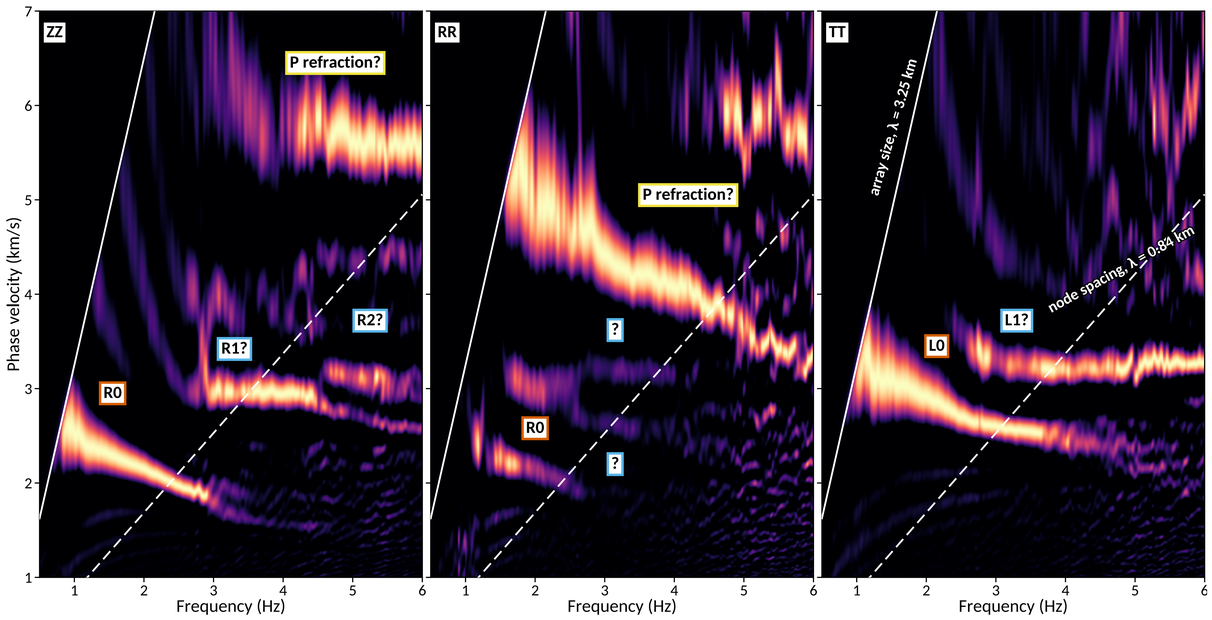
All-pairs offset gathers, 0.6–4.5 Hz, common clip
Rows: source; columns: receiver



Narrow-band gathers; lines: measured envelope moveout of fundamental (solid orange), overtone (dashed blue) and refraction-candidate (dotted yellow) arrivals

DISPERSION AND MODES

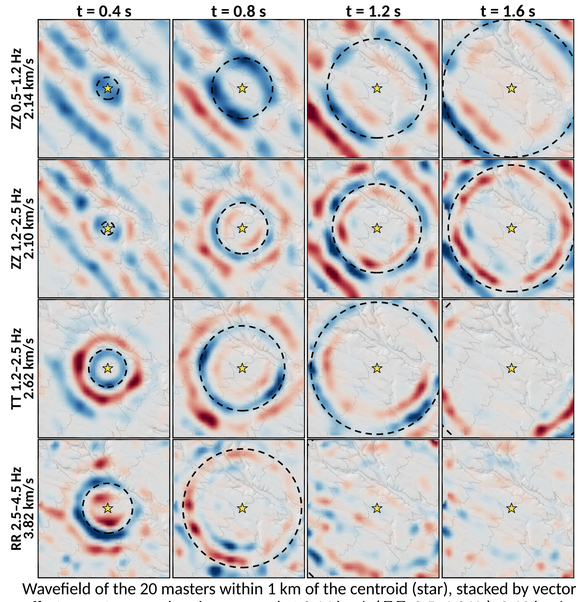
- Normal dispersion on Rayleigh and Love: V_S increases with depth
- At least one higher-mode Rayleigh branch on RR
- Love slightly faster than Rayleigh



Frequency-Bessel stacks of the all-pairs gathers, scaled per frequency; branches marked "?" are candidates; solid white line: array-size limit, λ = 3.25 km, longer wavelengths masked; dashed white line: node-spacing limit, λ = 0.84 km, shorter wavelengths can alias

TOPOGRAPHY VS. HETEROGENEITY

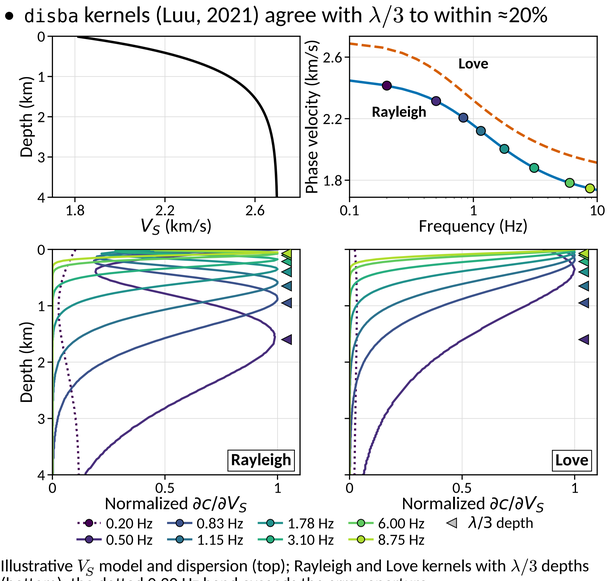
- Wavelengths **0.5–3.5 km** ≫ **170 m** relief: topography is sub-wavelength
- Wavefronts cross the array undeflected
- Rayleigh–Love cross-term leakage rises from **0.34** (0.1–0.3 Hz) to **0.90** (5–8 Hz)
- Departures from 1-D: lateral velocity heterogeneity in the shallow section
- Linear bands: Rayleigh energy from the SW, 2.2–2.5 km/s, confirmed by beamforming



Wavefield of the 20 masters within 1 km of the centroid (star), stacked by vector offset, over topography; rings expand at 2.14 km/s (ZZ, 0.5–1.2 Hz), 2.10 km/s (ZZ, 1.2–2.5 Hz), 2.62 km/s (TT) and 3.82 km/s (RR); dashed: measured crest

DEPTH SENSITIVITY

- Wavelength scaling: $z \approx \lambda/3 = c/(3f)$
- Sensitivity from **≈70 m to 1 km**
- `disba` kernels (Luu, 2021) agree with $\lambda/3$ to within ≈20%



Illustrative V_S model and dispersion (top); Rayleigh and Love kernels with $\lambda/3$ depths (bottom); the dotted 0.20 Hz band exceeds the array aperture

CONCLUSIONS

- Surface-wave-dominated 9-C field: Rayleigh on $\{ZZ, ZR, RZ, RR\}$, Love on TT
- Normal dispersion, with at least one higher mode on both Rayleigh and Love
- Sensitivity from 70 m to 1 km depth
- Topography is sub-wavelength; departures from 1-D record shallow lateral velocity heterogeneity

FUTURE WORK

- Rayleigh + Love branches: constrain a fully elastic model
- Ambient wavefield tomography (AWT) and eikonal tomography for lateral structure
- I-FDG (Davis et al., 2026, *GJI*) and HVSR for vertical constraints
- Higher modes add constraints to AWT and I-FDG
- Trans-dimensional Bayesian inversion: AWT, eikonal, I-FDG and HVSR constrain each other, giving $V_S(z)$ with uncertainty
- Pre-injection baseline for time-lapse monitoring